Off-Grid **U**tility for **T**hermal **A**irflow and **G**eneral **E**ase (**OUTAGE**): **A Cooling Device for Blackout Scenarios**

SUMMARY

The Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) addresses the critical issue of thermal discomfort during power outages, particularly in underserved communities. This innovative system integrates solar panels, rechargeable batteries, and a refillable water/ice reservoir to power a micro air pump, employing an energy-efficient evaporative cooling mechanism. Its portable and adaptable design enables deployment in high-risk areas, effectively overcoming the limitations of traditional cooling solutions.

A key innovation of OUTAGE lies in its dual power system, which combines solar energy harvesting with battery storage to ensure uninterrupted operation, even during extended blackouts. This sustainable approach reduces reliance on grid-based power, minimizes energy consumption, and aligns with global efforts to mitigate climate change through renewable energy adoption.

In summary, OUTAGE presents a significant contribution to off-grid cooling technologies, offering a scalable, environmentally sustainable, and cost-effective solution to address thermal discomfort. Its innovative design and sustainable operation enhance energy resilience, improve living conditions, and support global climate action objectives.

BACKGROUND AND THE PROBLEM

Outage, an interruption of electricity, disrupts daily life and critical operations across various regions, highlighting the urgent need for robust solutions in energy distribution, infrastructure resilience, and emergency preparedness.

According to the World Health Organization (WHO, 2022), heat-related illnesses are on the rise, particularly in India, Africa, and parts of Southeast Asia, with poor access to cooling solutions.

To address frequent power outages and high electricity costs, the Philippine government has implemented several initiatives to enhance energy resilience and accessibility. The Energy Resiliency and Contingency Plan was introduced to modernize energy infrastructure and reduce power interruptions nationwide (National Grid Corporation of the Philippines, 2020). Additionally, the Household Electrification Program aims to bring electricity to outage and underserved areas, ensuring more communities have reliable power during blackouts (DOE, 2020). Despite these efforts, challenges persist in providing affordable and sustainable cooling solutions, especially in rural areas where power infrastructure remains underdeveloped.

In Davao Oriental, frequent power outages during the dry season worsen the discomfort caused by high temperatures. Limited access to electricity infrastructure in rural areas leaves many residents without reliable cooling systems, making alternative, outage solutions crucial for coping with the heat (Davao Oriental Provincial Government, 2021).

Given this outlook, the researchers are driven to develop the Off-grid Utility for Thermal Airflow and General Ease (OUTAGE), a timely and relevant innovation designed to address the discomfort caused by high temperatures during blackouts.

This study presents a promising solution to the challenges of maintaining thermal comfort without relying on the traditional power grid.

As the proposed intervention, OUTAGE offers an adaptable and cost-effective cooling solution by utilizing recycled materials and a low-energy setup. By ensuring accessibility and sustainability, the device aims to improve indoor and outdoor comfort during power outages, particularly in underserved communities, thereby contributing to better living conditions and energy resilience.

BENEFICIARIES

The Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) represents an economically viable and environmentally sustainable approach to enhancing comfort during power interruptions. This innovative device aligns with the United Nations Sustainable Development Goals (SDG 7: Affordable and Clean Energy, SDG 11: Sustainable Cities and Communities, and SDG 13: Climate Action). With over 940 million people globally lacking access to reliable electricity (IEA, 2022), OUTAGE seeks to improve quality of life, reduce ecological footprints, and bolster resilience to climate-related challenges. Below are its key beneficiaries:

Households in Rural and Underserved Areas. In the Philippines, 12% of the population lacks access to electricity, with rural regions experiencing the most frequent power outages (World Bank, 2022). During these blackouts, heat-related illnesses like heat stroke can increase by 20% (WHO, 2022). OUTAGE offers these households an affordable cooling solution, reducing heat-related health risks and enhancing living conditions, particularly during the dry season when temperatures can reach 40°C.

Environmental Advocates and Sustainability Enthusiasts. Globally, cooling systems account for 10% of total electricity consumption (IEA, 2021). By using recyclable materials and reducing dependency on energy-intensive systems, OUTAGE supports environmental advocates in addressing climate change. Additionally, its use of renewable solar energy helps reduce CO_2 emissions, contributing to the goal of limiting global warming to 1.5°C as outlined in the Paris Agreement.

Small Business Owners. Power outages cost small businesses in the Philippines an estimated ₱30,000 per year in lost sales and spoiled goods (DOE, 2022). For example, perishable goods in sari-sari stores spoil within hours during blackouts. OUTAGE provides a cost-effective cooling alternative, ensuring customer comfort and product preservation, helping reduce financial losses by up to 15% annually.

Construction and Outdoor Workers. In the Philippines, heat stress contributes to a 5% decline in worker productivity, particularly in outdoor sectors like construction (ILO, 2022). High temperatures, which often exceed 35°C, pose serious health risks, including dehydration and heat exhaustion. OUTAGE offers a portable, energy-efficient cooling system that mitigates these risks, improving worker safety and maintaining productivity levels.

OUTAGE exemplifies a practical, sustainable, and economically viable strategy for mitigating thermal discomfort during blackouts. By addressing the needs of rural households, small businesses, environmental advocates, and outdoor workers, OUTAGE promotes energy efficiency and fosters climate resilience. With its alignment to SDGs 7, 11, and 13, OUTAGE is a step toward creating sustainable and inclusive communities while addressing the growing challenges of climate change and energy accessibility.

THE PROPOSED SOLUTION FOR THE PROBLEM

The Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) will be developed. This innovative system will provide a sustainable thermal relief without relying on energy-intensive technologies, bridging the gap between energy accessibility and climate resilience.

OUTAGE incorporates scientifically proven passive cooling strategies, such as evaporative cooling, to lower ambient temperatures. By utilizing a refillable water/ice reservoir, the device cools incoming air through evaporation, reducing indoor and outdoor temperatures during power outages. Its lightweight, portable modules ensure immediate relief for households, small businesses, and outdoor workers, offering a practical alternative to traditional cooling methods.

The system uses a dual power source: lightweight solar panels and rechargeable battery storage. Solar energy is harvested during the day, providing continuous cooling while simultaneously charging the built-in batteries. This ensures uninterrupted operation at night or during prolonged power outages, aligning with renewable energy goals and reducing dependence on fossil fuel-based grid power. This approach promotes energy resilience and lowers greenhouse gas emissions, in line with global climate mitigation strategies.

A key strength of OUTAGE lies in its adaptability across various environments. Unlike conventional cooling systems, its compact and deployable design allows strategic placement in high-risk areas, making it suitable for urban centers, remote rural communities, and mobile workspaces such as construction sites.

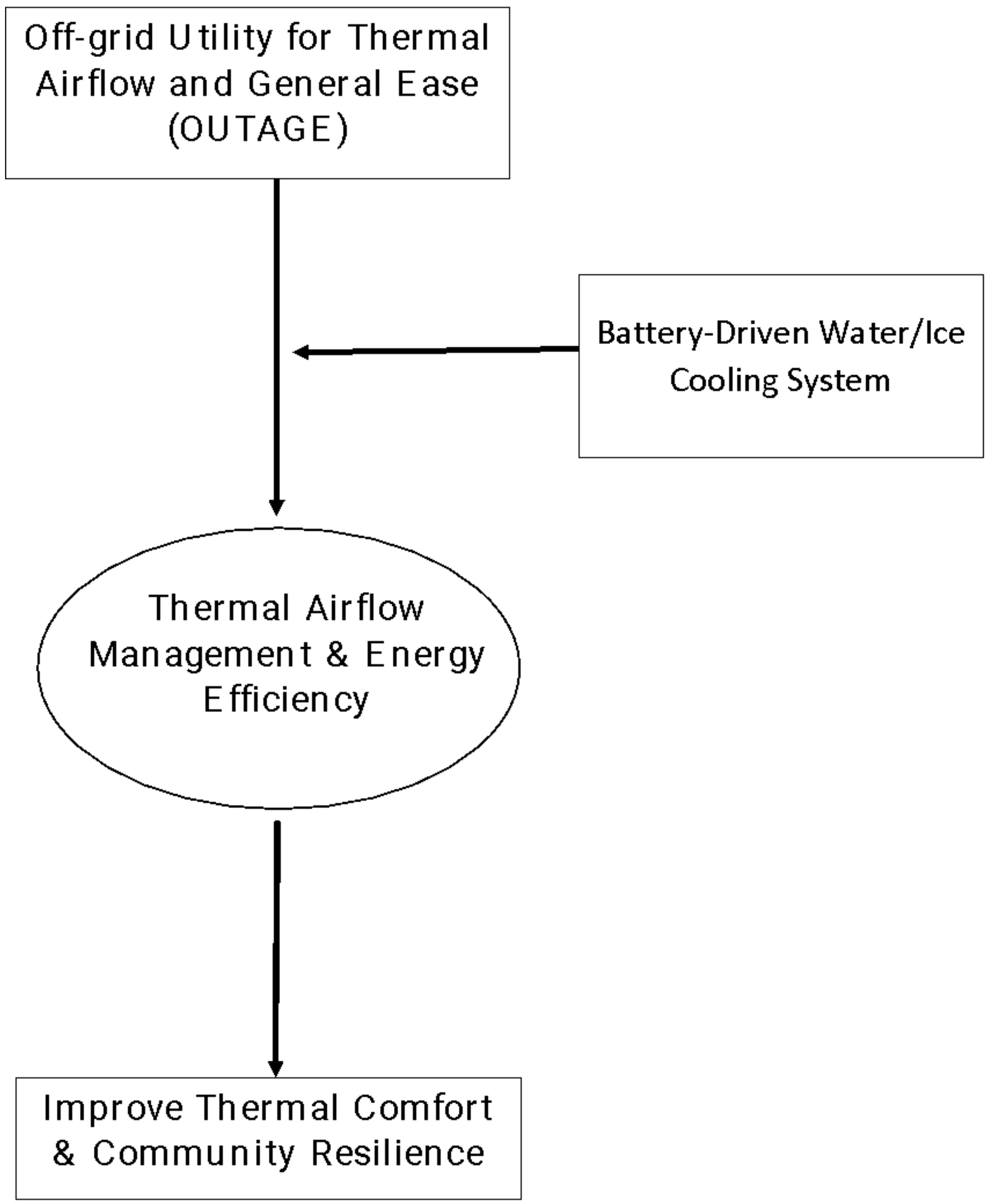
Furthermore, OUTAGE enhances energy resilience by reducing dependence on grid-powered cooling solutions, aligning with global efforts to combat climate change. By offering an affordable and efficient alternative, it supports the well-being of vulnerable populations while contributing to sustainable energy practices.

The device is designed using advanced computational fluid dynamics (CFD) simulations to optimize airflow patterns and enhance the thermodynamic efficiency of the cooling process. The airflow system ensures even distribution of cool air, while the high-efficiency micro air pump delivers targeted airflow with minimal energy usage. Scientific testing under controlled conditions will validate its thermal performance, ensuring a consistent cooling effect across diverse environments.

In addition to providing thermal comfort, OUTAGE fosters environmental sustainability by promoting the use of low-energy, eco-friendly technologies. Its implementation not only addresses immediate cooling needs but also strengthens the long-term climate resilience of affected communities.

The Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) offers an innovative and sustainable solution to mitigate the impacts of power outages and extreme heat. Through its adaptability and commitment to sustainability, OUTAGE improves the quality of life for vulnerable populations while supporting global climate objectives.

Figure 1: The Conceptual Framework of the Proposed Solution



METHODS / DETAILS OF THE PROPOSED SOLUTION

The Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) integrates innovative features and sustainable practices to provide efficient cooling during power outages. The proposed solution follows a systematic approach, integrating various components and methods to ensure its effectiveness. Below is a detailed breakdown of the methods and technical steps involved in its development and implementation:

1. Dual Power Supply: Solar and Electric Charging Systems

a. Incorporate a compact solar panel system to harness renewable energy during daylight hours.

b. Use a DC-DC converter to stabilize the voltage output from the solar panels for efficient energy transfer to the battery.

c. Select lithium-ion batteries for their high energy density and reliability.

d. Include a battery management system (BMS) to monitor and regulate charging cycles, preventing overcharging and extending battery life.

2. Micro Air Pump Integration for Rapid Cooling

a. Choose a low-power micro air pump capable of delivering sufficient airflow to cool a small room.

b. Integrate the pump with an adjustable fan speed controller to allow users to regulate airflow based on their cooling needs.

c. Optimize the pump's operation to balance rapid cooling with minimal energy consumption.

3. Integrated Refillable Water/Ice Reservoir

a. Large-capacity reservoir designed to hold water or ice serves as the core cooling element.

b. Air is drawn through the reservoir, where it picks up the cool temperature of the water or ice before being circulated.

c. The reservoir includes a top-opening refill mechanism, enabling users to easily add water or ice without disassembling the device.

4. Recycled Material Utilization

a. Partner with local recycling facilities to source durable, lightweight recycled materials such as aluminum, plastic, and cardboard.

b. Conduct a life cycle analysis to evaluate the environmental benefits of using recycled materials.

5. Portable and Lightweight Design

a. Design the device with built-in handles and a compact form factor for easy transportation.

b. Ensure the weight does not exceed 5 kg for portability by users of varying ages and physical abilities.

c. Conduct drop and vibration tests to ensure the device can withstand frequent movement and rough handling.

6. Passive Cooling Implementation

a. Ventilation Optimization

Include strategically positioned vents to enhance natural airflow.

b. Thermal Insulation

Integrate reflective and insulating materials in the casing to reduce heat absorption.

7. Energy Storage and Management

a. Incorporate high-capacity batteries with a minimum storage capacity of 20000 mAh to ensure prolonged operation.

b. Perform endurance tests to evaluate battery runtime under continuous operation and varying cooling levels.

8. User-Friendly Controls and Monitoring

a. Develop a simple interface with buttons for power on/off, fan speed adjustment, and mode selection (solar or battery).

b. Include LED indicators for battery status, charging progress, and system faults to keep users informed of the device's operational status.

c. Test the interface with target users to ensure accessibility and intuitive operation.

9. Environmental and Economic Impact Assessment

a. Calculate the reduction in greenhouse gas emissions due to the use of solar power and recycled materials.

b. Compare the manufacturing and operational costs of OUTAGE to traditional cooling systems.

c. Highlight long-term savings in energy costs for end-users.

10. Deployment and Maintenance Strategy

a. Identify and prioritize high-risk areas with frequent power outages for initial deployment.

b. Collaborate with local governments and non-profit organizations to facilitate distribution.

c. Train local technicians on device maintenance and repairs.

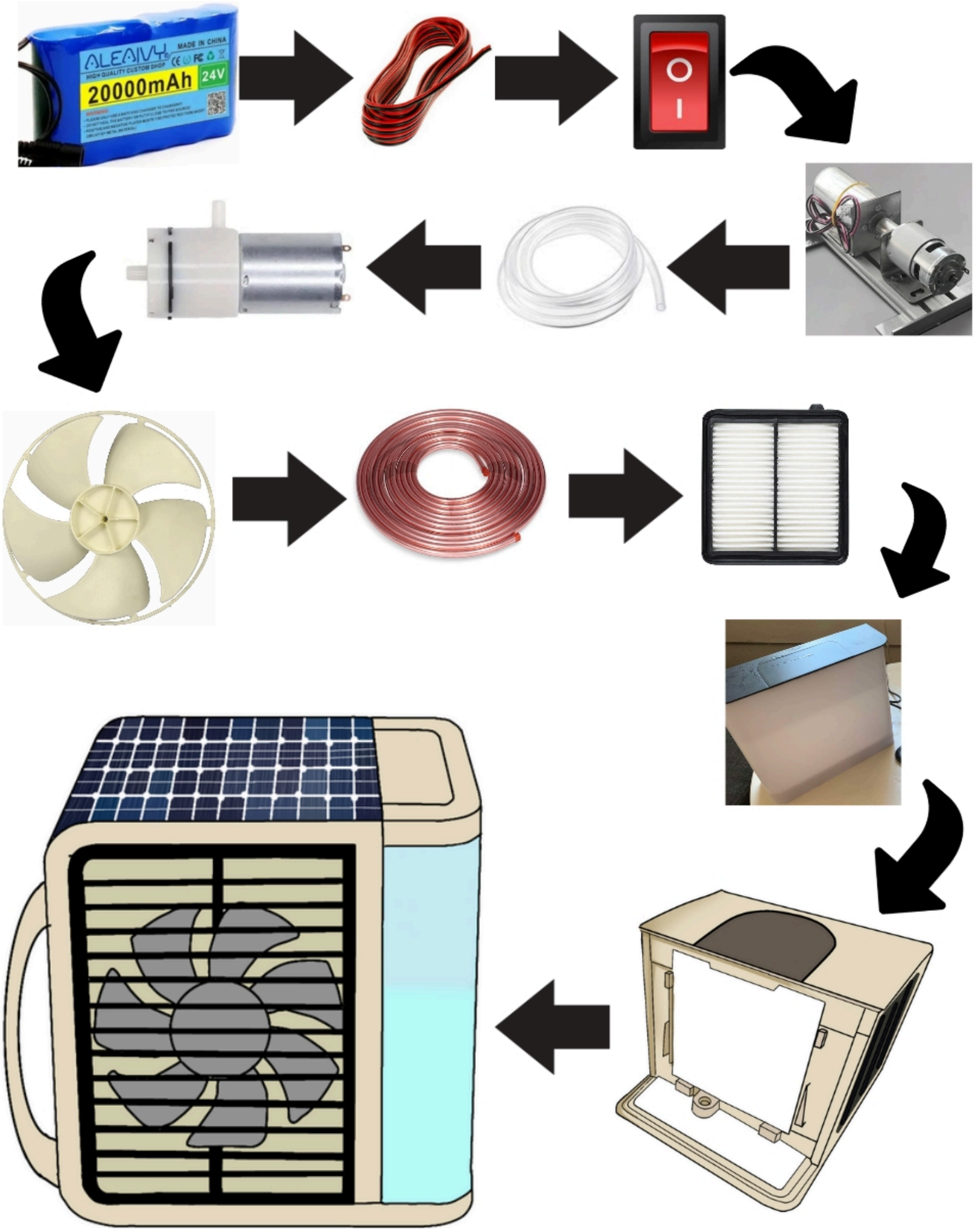
11. Safety and Compliance Testing

a. Ensure compliance with international safety standards, including electrical and thermal safety.

b. Obtain certifications such as CE, RoHS, or local equivalent to guarantee safety and reliability.

By implementing these methods, OUTAGE will deliver a reliable, sustainable, and user-friendly solution to combat thermal discomfort during blackouts. This comprehensive approach ensures that the device meets the needs of vulnerable communities while promoting environmental responsibility and energy resilience.

Figure 2: Flow Chart of the Proposed Solution



The flow chart illustrates in Figure 2 are the functions and process of Off-Grid Utility for Thermal Airflow and General Ease (OUTAGE) which consists of several key components, each serving a specific function to provide efficient cooling during power outages.

Energy Harvesting

OUTAGE integrates both solar panels and rechargeable batteries to harvest and store energy. The solar panels are optimized to capture sunlight efficiently, while the batteries provide a backup power source during low-light conditions. This dual energy system ensures continuous operation and makes OUTAGE suitable for deployment in diverse environments, from urban to rural areas.

Powering the System

The collected solar energy and stored battery power drive the entire cooling mechanism of OUTAGE. This ensures a sustainable, autonomous operation without reliance on the electrical grid. The system's energy-efficient design extends its usability, making it ideal for off-grid scenarios.

Cooling Mechanism

The device employs a micro air pump that circulates air over water or ice within a refillable reservoir. This process creates a cooling effect through evaporative cooling, effectively reducing ambient temperatures. The lightweight pump and its low-energy requirements make the system both effective and economical.

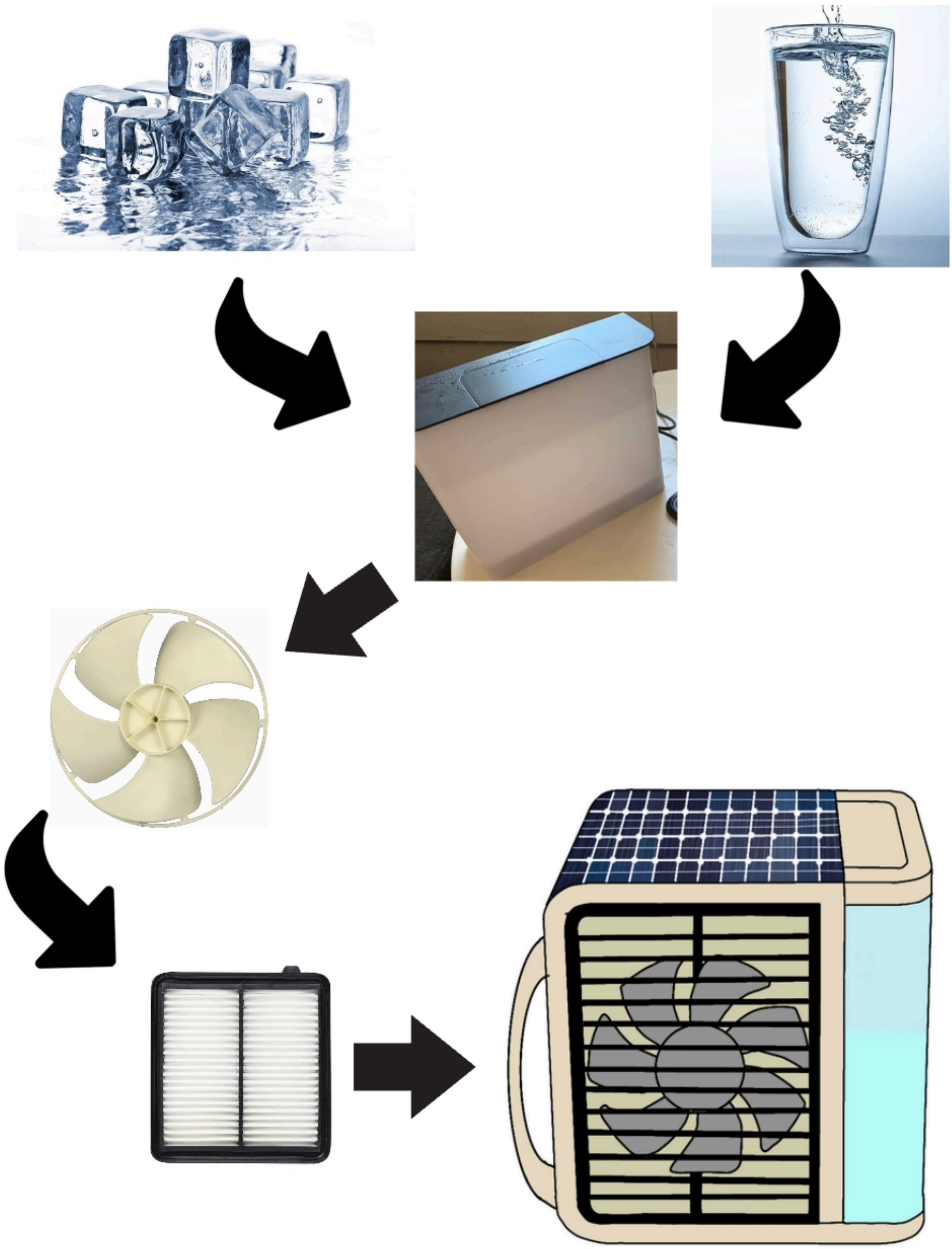
Refillable Water/Ice Reservoir

The system includes a refillable reservoir for water or ice, enabling flexible and consistent cooling. Users can quickly replenish the reservoir, ensuring the device remains functional during extended periods of use.

Adaptive Deployment System

The portable design of OUTAGE allows it to be strategically placed in high-heat-risk zones, such as households, small businesses, and construction sites. Its adaptability ensures that vulnerable populations can access cooling solutions tailored to their specific needs.

Figure 3: Flow Chart in Processing the Water or Ice to Cooling Airflow Circulation System



The flow chart in Figure 3 shows the process of turning water or ice into a cooling airflow system to combat thermal discomfort during blackouts.

Water or Ice Reservoir Filling

The cooling process begins by filling the designated reservoir with water, ice, or a combination of both. This reservoir acts as the primary source for the cooling mechanism, providing a thermal medium to reduce the surrounding temperature.

Activation of the Micro Air Pump

Once the reservoir is filled, the micro air pump is activated. The pump functions as the core component, drawing ambient air into the system and pushing it over the surface of the water or ice.

Air Circulation and Evaporation

As the air flows over the water or ice, heat energy from the air is absorbed, causing the water to evaporate slightly or the ice to melt. This phase change results in a reduction of air temperature, creating a cooling effect.

Release of Cooled Air

The cooled air is then expelled through the system's ventilation outlets, providing a refreshing airflow to the immediate surroundings. The lightweight design ensures even distribution of the cooled air across the intended area.

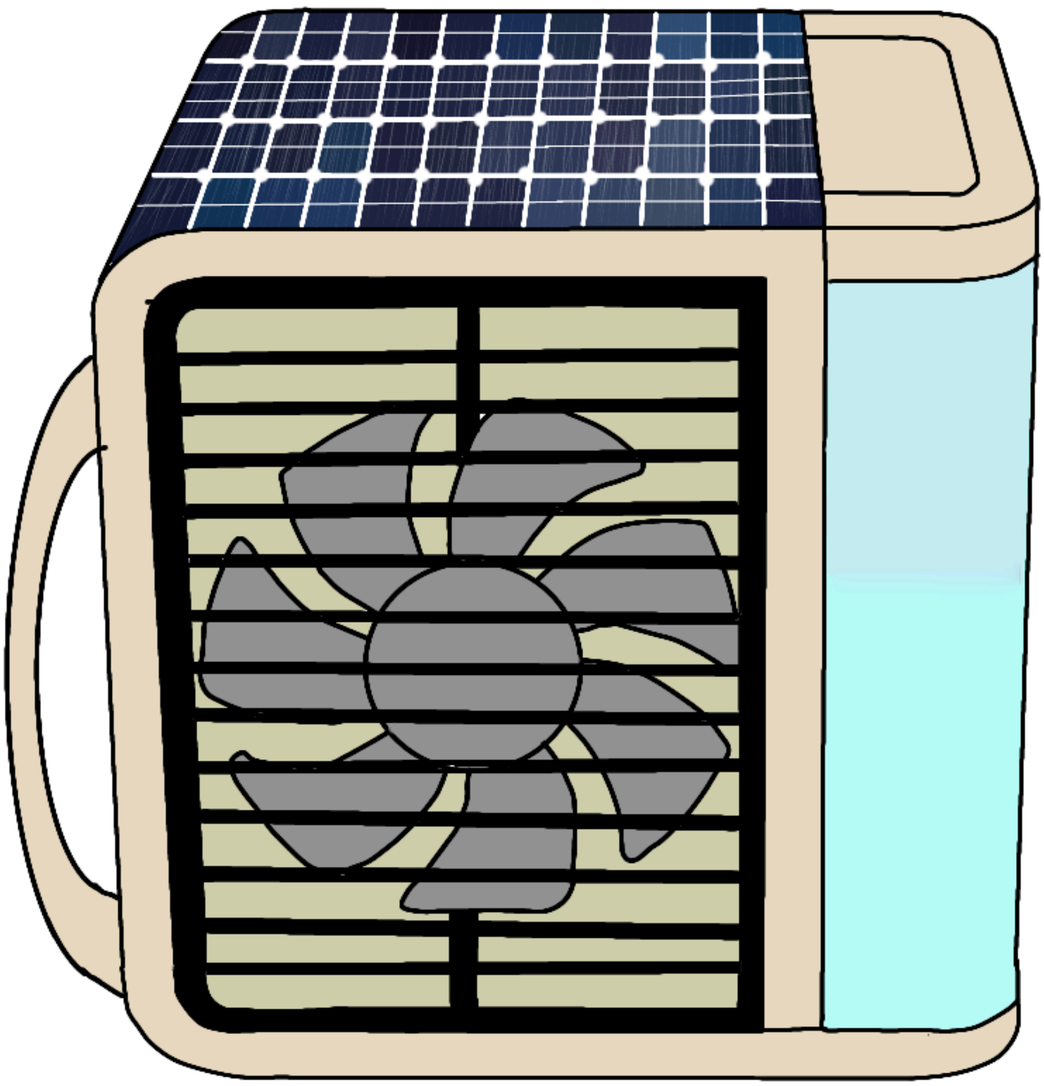
Continuous Operation

The process repeats in a continuous loop, ensuring consistent cooling as long as water or ice remains in the reservoir. The low-energy pump operates efficiently, maximizing thermal relief without requiring significant power input.

Reservoir Maintenance and Refilling

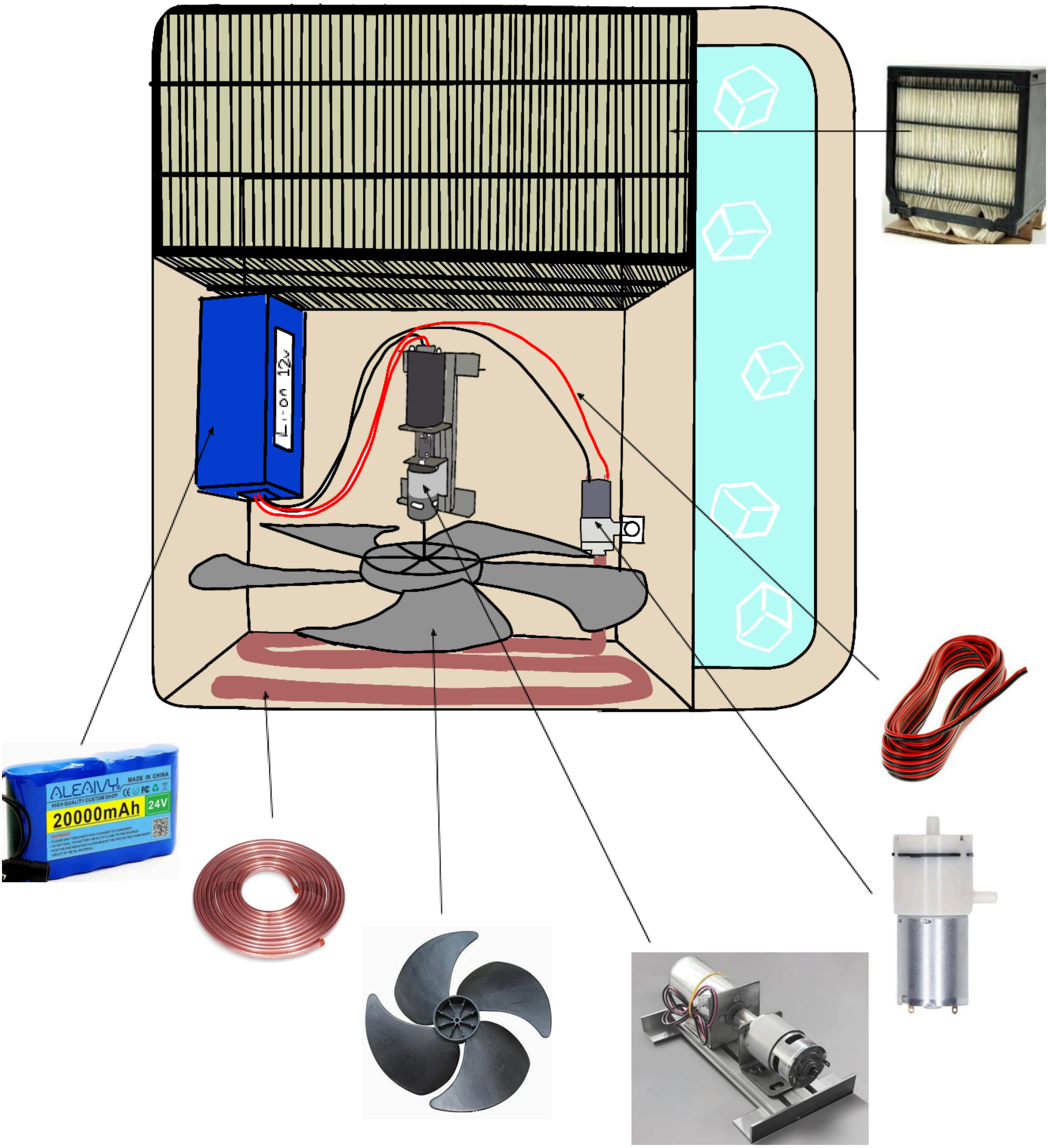
Once the water has fully evaporated or the ice has melted, the reservoir can be easily refilled to resume cooling. This simple and refillable design ensures sustained operation during extended blackouts or high-temperature periods.

Figure 4: The External Prototype of the Proposed Solution



A compact and portable cooling device designed to alleviate thermal discomfort during power outages. Featuring an integrated solar panel, rechargeable battery, and refillable water or ice reservoir, OUTAGE powers a micro air-pump that utilizes energy- efficient evaporative cooling.

Figure 4: The Internal Prototype of the Proposed Solution



Solar and Electric Charging System

OUTAGE features a dual power system that combines solar panels and rechargeable batteries, ensuring continuous operation during power interruptions. Solar panels provide renewable energy during the day, while the battery system stores energy for nighttime or prolonged blackouts, ensuring sustainability and reliability in underserved areas.

Micro Air Pump with Refillable Cooling System

The micro air pump works in tandem with a refillable water or ice chamber to enhance cooling efficiency. The pump circulates air over the water or ice, delivering a cooling effect that is both energy-efficient and effective. Users can easily refill the chamber, ensuring consistent cooling during extended blackouts or extreme heat conditions.

Recycled Material Construction

Constructed with recycled materials such as aluminum and plastic, OUTAGE demonstrates a commitment to environmental sustainability. This approach reduces production costs while maintaining durability and functionality, supporting eco-friendly practices.

Portable and Lightweight Design

The device's compact and lightweight design ensures easy transportation and deployment across various environments. Its portability makes it particularly suitable for households, outdoor workers, and small businesses in remote or disaster-prone areas.

Smart Control Interface

The device is equipped with a user-friendly control panel, allowing adjustments to fan speed and power mode. LED indicators provide real-time feedback on battery levels, solar input, and system status, simplifying operation and maintenance.

Adaptable Deployment System

This design allows for strategic placement in a variety of settings, from urban homes to rural communities and outdoor worksites. Its adaptability ensures optimal airflow and cooling, even in poorly ventilated areas.

Energy-Efficient Operation

Combines passive cooling techniques with active low-energy systems to maximize cooling efficiency while minimizing energy consumption. This ensures prolonged operation during blackouts and supports long-term sustainability.

Continuous Improvement

OUTAGE has been developed with a focus on continuous improvement. By systematically monitoring the system's performance and incorporating user feedback, enhancements will be implemented to increase the device's efficiency, adaptability, and overall effectiveness in delivering thermal relief. These iterative improvements will ensure that OUTAGE is capable of evolving to address the diverse needs of various environments and user requirements, thereby optimizing its effectiveness in alleviating heat-related issues during power outages.

The cost analysis presented in Table 1 offers a comprehensive overview of the materials and associated expenses necessary for the proposed solution. This analysis encompasses all critical components and activities required for the system's implementation, thereby providing a clear understanding of the financial implications involved.

The total expenditure for all materials needed for the proposed solution is ₱1,175. This cost analysis serves as a significant tool for budget planning and resource management, enabling stakeholders to allocate financial resources efficiently and facilitate the successful implementation of OUTAGE.

Table 1: Cost-Analysis of the Proposed Solution

ACTIVITIES	MATERIALS	UNIT	PRICE	QUANTITY	TOTAL
Body Structure	Air Pump	pc	₱130	1	₱130
	Battery	pc	₱650	1	₱650
	Solar Panel	pc	₱170	1	₱170
	Motor	pc	₱150	1	₱150
	Fan	pc	₱75	1	₱75
TOTAL					**₱1,175**

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